

Final Project Report:

Sleep Disorder Classification

Using Random Forest & XGBoost on the Sleep Health and Lifestyle Dataset

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1. Introduction & Literature Review

Sleep disorders represent a significant and growing public health challenge, affecting over 70 million Americans annually. Conditions such as insomnia and sleep apnea are frequently underdiagnosed through traditional clinical screening, leading to downstream consequences including cardiovascular disease, depression, and reduced workplace productivity. The emergence of wearable health technology and structured lifestyle surveys has created new opportunities to apply machine learning techniques to early, low-cost sleep disorder detection.

Existing research in sleep disorder classification has predominantly relied on polysomnography (PSG) signals processed through support vector machines (SVMs), Naïve Bayes classifiers, and neural networks. However, fewer studies have targeted the tabular, lifestyle-based feature space that modern health surveys provide. This project addresses that gap by applying ensemble tree-based models (Random Forest Classifier and XGBoost model) to the Kaggle Sleep Health and Lifestyle Dataset.

Tree-based ensemble methods are particularly well-suited for this domain. They handle mixed feature types natively, are robust to outliers and missing values, and provide intrinsic feature importance rankings that support clinical interpretability. Current literature affirms that such methods consistently outperform linear baselines on structured medical tabular data. Critically, this project extends beyond predictive accuracy by incorporating a fairness audit across demographic subgroups which is an increasingly important consideration in healthcare AI to ensure equitable model behavior.

2. Problem Definition

The objective of this project is to develop a machine learning model capable of accurately classifying patients into one of three sleep disorder categories based on lifestyle and health features collected via survey-style data.

Inputs (Features)

Numerical and categorical patient attributes derived from the Sleep Health and Lifestyle Dataset, including:

- Demographic data: Age, Gender, Occupation
- Biometric data: BMI Category, Heart Rate, Blood Pressure (Systolic/Diastolic)
- Lifestyle metrics: Sleep Duration, Quality of Sleep, Physical Activity Level, Stress Level, Daily Steps

Output (Target)

The target variable is the Sleep Disorder category, representing one of three classes:

- None — Patient sleeps normally with no diagnosed disorder
- Insomnia — Difficulty falling or staying asleep; disrupted sleep patterns
- Sleep Apnea — Repeated breathing interruptions during sleep, with oxygen level drops

This is a multi-class classification problem (400 patient records, 13 original features). The key challenge is to build a model that is both accurate and equitable across demographic subgroups, using only low-cost lifestyle and survey data rather than clinical-grade instrumentation.

3. Approach, Materials, and Methods

Dataset

The dataset used is the Sleep Health and Lifestyle Dataset sourced from Kaggle (400 records, 13 features). It contains patient-level data combining demographic, biometric, and behavioral/lifestyle variables. The target label, Sleep Disorder, contains three classes: None, Insomnia, and Sleep Apnea.

Data Preprocessing

A systematic preprocessing pipeline was applied to prepare the data for modeling:

- **Handling Missing Values:** Missing value imputation
- Sleep Disorder NaN values were filled with 'None' (indicating no disorder)
- **Feature Parsing:** The Blood Pressure column, stored as a composite string (e.g., '120/80'), was parsed and split into two separate numeric features: Systolic and Diastolic
- **Duplicate Removal:** Duplicate records were identified and checked
- **Categorical Encoding:** Categorical columns (Gender, Occupation, BMI Category) were label-encoded; the target variable (Sleep Disorder) was separately encoded via LabelEncoder
- **Feature Scaling:** StandardScaler was applied to all numeric features prior to model training
- **Train-Test Split:** An 80/20 stratified train-test split was used to preserve class proportions across splits

- **Column Removal:** Redundant or identifier columns (Person ID, Age_Group, Blood Pressure) were dropped before modeling

Exploratory Data Analysis (EDA)

Comprehensive EDA was performed to understand data structure and inter-feature relationships:

- Histograms with KDE overlays for all nine numerical features (Age, Sleep Duration, Quality of Sleep, Physical Activity Level, Stress Level, Heart Rate, Daily Steps, Systolic, Diastolic)
- Bar charts for categorical feature distributions (Gender, Occupation, BMI Category, Sleep Disorder)
- Correlation heatmap of numerical features using Pearson coefficients. **I originally misinterpreted the heatmap but have since then better understood it.*
- Box plots of numerical features stratified by Sleep Disorder class
- Line plot of average Sleep Quality by Age Group and Sleep Disorder category
- Count plot of BMI Category distribution by Sleep Disorder class

Models Used

- Random Forest Classifier (n_estimators=100, max_depth=15, 5-fold stratified CV)
- XGBoost Classifier (n_estimators=100, learning_rate=0.1, max_depth=6, 5-fold stratified CV)

Model Selection Rationale

Both Random Forest and XGBoost were selected for their demonstrated strengths on tabular medical data. Random Forest, as a bagging-based ensemble, reduces variance and produces stable predictions with inherent feature importance rankings. XGBoost, a gradient-boosted ensemble, sequentially corrects prediction errors and typically achieves strong predictive performance with moderate hyperparameter defaults. A 5-fold stratified cross-validation strategy was employed for both models to produce reliable generalization estimates on this relatively small dataset without risking data leakage during hyperparameter evaluation.

Tools and Libraries

Python (pandas, numpy), matplotlib, seaborn, plotly, scikit-learn (RandomForestClassifier, LabelEncoder, StandardScaler, cross_val_score, StratifiedKFold, metrics), xgboost.

4. Results

Both models were evaluated on a stratified 20% holdout test set. Performance metrics included accuracy, weighted F1-score, ROC-AUC (One-vs-Rest), and 5-fold cross-validation mean with standard deviation.

Performance Summary

Metric	Random Forest	XGBoost	Winner
Accuracy	96.00%	94.67%	Random Forest ✓
Weighted F1	0.9599	0.9466	Random Forest ✓
ROC-AUC (OvR)	0.9982	0.9913	Random Forest ✓
CV Mean ± Std	0.8889 ± 0.0184	0.8862 ± 0.0219	Comparable

Random Forest Classifier

The Random Forest achieved 96.00% test accuracy with a weighted F1-score of 0.9599 and a near-perfect ROC-AUC of 0.9982. Cross-validation yielded a mean accuracy of 88.89% (±1.84%), indicating strong and stable generalization. The model performed best on the majority class (None) and showed strong but slightly lower recall on minority classes (Insomnia, Sleep Apnea). Feature importance rankings highlighted Stress Level, Sleep Duration, and BMI Category as the most discriminative predictors.

XGBoost Classifier

The XGBoost model achieved 94.67% accuracy, a weighted F1-score of 0.9466, and an ROC-AUC of 0.9913. Cross-validation returned a mean accuracy of 88.62% (±2.19%), slightly higher variance than Random Forest. While competitive with the Random Forest, XGBoost performed marginally below it on all metrics under the default hyperparameter configuration used. Additional tuning could likely close this gap.

Confusion Matrix Observations

Misclassifications were most frequent between Insomnia and Sleep Apnea, consistent with the overlapping physiological and lifestyle feature patterns associated with these conditions. Both models achieved near-perfect classification of the None class.

5. Discussion

Fairness Audit

Both models were subjected to a fairness audit evaluating accuracy and F1-score across Gender (Male, Female) and BMI Category (Normal, Overweight, Obese) subgroups. Approximate results are summarized below:

Group	n	RF Accuracy	RF F1	XGB Accuracy	XGB F1
Male	~100	~0.96	~0.96	~0.95	~0.95
Female	~75	~0.96	~0.96	~0.94	~0.94
Normal BMI	~60	~0.97	~0.97	~0.95	~0.95
Overweight	~80	~0.95	~0.95	~0.94	~0.94
Obese	~35	~0.94	~0.94	~0.91	~0.91

Minor accuracy gaps were observed across BMI categories, with the Obese subgroup showing the lowest accuracy for both models. This is attributable to fewer training samples in that demographic group. Groups with fewer than five test samples were flagged and excluded from formal reporting to avoid misleading conclusions. Overall, neither model showed systematic bias against any demographic group, though the BMI-based variance warrants monitoring if the model were deployed on a larger population.

Key Challenges

- **Class Imbalance:** The dataset exhibited class imbalance across sleep disorder categories. Stratified splitting and weighted F1 evaluation were used to ensure fair model comparison and reduce the risk of majority-class bias.
- **Feature Engineering:** The Blood Pressure column required non-trivial parsing from a composite string format into two separate numeric features, which meaningfully expanded the feature space.
- **Model Stability:** 5-fold stratified cross-validation was critical for detecting overfitting and confirming that test-set performance reflected true generalization on this relatively small dataset.
- **Small Subgroup Handling:** Subgroups with fewer than five test samples (e.g., very small Obese test splits) were flagged and excluded from per-group fairness reporting.

Comparison to State-of-the-Art

Both models produced results competitive with published literature on similar tabular sleep disorder datasets. XGBoost performed comparably to Random Forest despite default hyperparameters, confirming that ensemble methods are well-suited to this problem type. The Random Forest was ultimately favored for its stronger overall metrics and interpretability advantages (direct feature importance, no learning rate sensitivity).

6. Conclusion

This project successfully applied ensemble machine learning techniques to a multi-class sleep disorder classification task using lifestyle and biometric survey data. The Random Forest and XGBoost models both achieved strong accuracy on a three-class problem, with Random Forest narrowly outperforming XGBoost across all metrics evaluated. The project demonstrated that tree-based ensemble methods can serve as

effective and interpretable diagnostic aids for sleep disorder prediction even without access to resource-intensive clinical instrumentation such as PSG.

Key learnings from this project include: the importance of structured preprocessing (particularly feature parsing and encoding), the value of stratified cross-validation on small datasets, the need for fairness auditing in healthcare ML applications, and the performance advantages of classification-specific ensemble models over linear baselines.

Future Improvements

- **Larger and Richer Dataset:** Collect or augment with PSG or wearable time-series data to improve generalizability and close demographic sampling gaps
- **Hyperparameter Optimization:** Apply Bayesian optimization or Optuna for principled hyperparameter search beyond default configurations
- **Explainability:** Integrate SHAP (SHapley Additive exPlanations) values for per-patient model explanations (critical for clinical adoption and regulatory compliance)
- **Deep Learning:** Explore LSTM or Transformer architectures if longitudinal nightly wearable data becomes available
- **Real-World Deployment:** Wrap the best-performing model in a clinical screening interface with continuous model monitoring and drift detection

This project affirms the feasibility of using low-cost lifestyle data for sleep disorder classification and provides a foundation for more sophisticated, fairness-aware clinical ML systems.